Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	22 October 2026
Team ID	SWUID2026403869
Project Name	IMovies – Movie Ticket Booking Website
Maximum Marks	4 Marks

Technical Architecture:

The IMovies platform is built using the MERN stack (MongoDB, Express.js, React.js and Node.js) and follows a scalable 3-tier architecture that includes the presentation layer (frontend), business logic (backend), and data storage layer. The solution ensures performance, security, and ease of integration with third-party APIs (e.g., Razorpay for payments).

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	React-based responsive web interface for customers to browse movies, search and filter, select shows and seats, make payments and view tickets	HTML, CSS, JavaScript, React.js, Bootstrap / Tailwind CSS
2	Application Logic-1	User registration and login, movie listing and search, show and seat selection, booking, cancellation and refund	Node.js, Express.js (REST APIs), JWT
3	Application Logic-2	Admin panel (movies, theatres, shows, coupons, reports), e-ticket generation with QR code, email/SMS notifications	React.js, Node.js, Nodemailer, QR Code library
4	Database	NoSQL database storing users, movies, theatres, shows, seats, bookings, payments and reviews (accessed through Mongoose)	MongoDB, Mongoose
5	Payment Gateway	Secure online payment through UPI, cards and net banking, and refund processing	Razorpay API
6	Cloud / Hosting	Hosting of frontend and backend, and cloud database	Vercel / Render, MongoDB Atlas

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	MERN stack: MongoDB (database), Express.js and Node.js (backend), React.js (frontend), with styling libraries	MongoDB, Express.js, React.js, Node.js, Bootstrap, Tailwind CSS
2	Scalable Architecture	3-tier architecture with RESTful APIs	MVC
3	Security Implementation	Secure login, password protection and encrypted data transfer	JWT authentication, bcrypt, HTTPS
4	Third-Party Integration	Payment processing and ticket notifications through external APIs	Razorpay API, Email / SMS service

References:

React.js Documentation

Node.js Introduction

Express.js Documentation

MongoDB Documentation

Razorpay Documentation

How to draw useful technical architecture diagrams